Serverless Media Processing Pipeline on AWS – Implementation Guide

1. Introduction

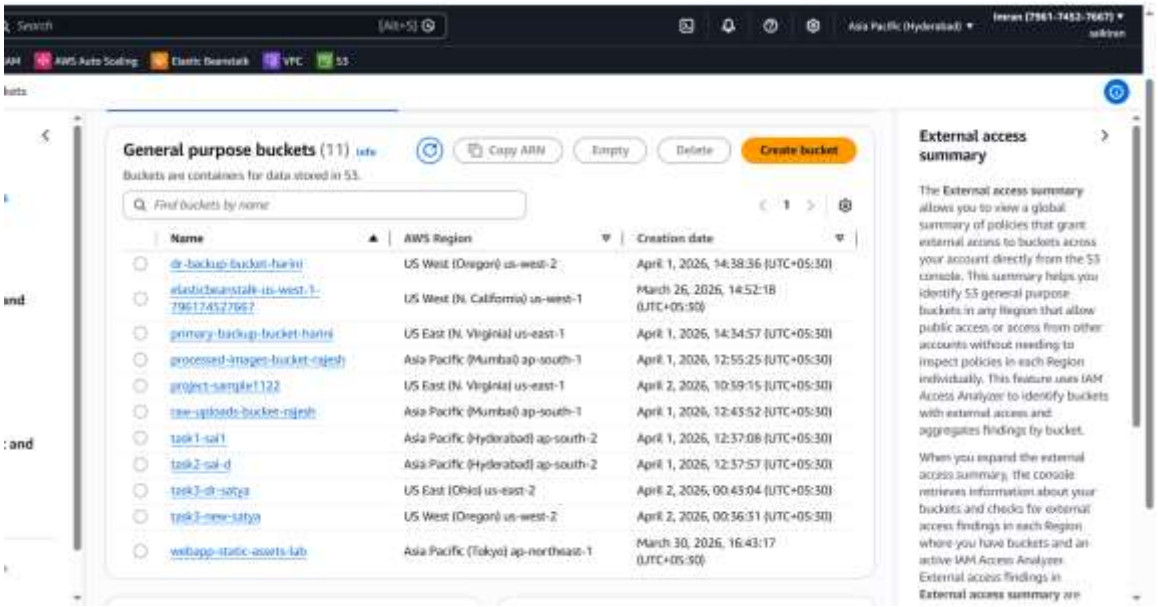
This document explains the step-by-step implementation of a serverless media processing pipeline using AWS services. It automates image uploads, processing, delivery, and storage optimization.

2. Architecture Overview

Browser → Pre-Signed URL → S3 (Raw Uploads)
→ Lambda (Resize + Watermark)
→ S3 (Processed Images)
→ CloudFront (+ Lambda@Edge)
→ End Users

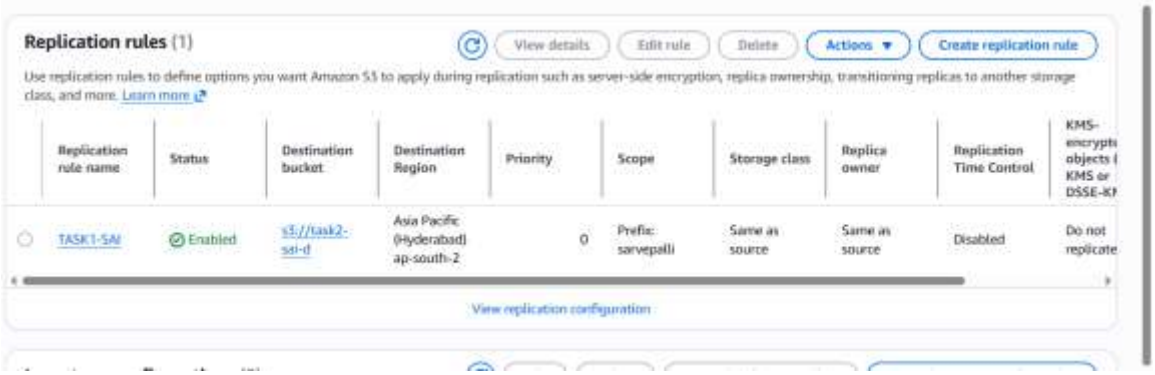
3. S3 Bucket Setup

Create raw-uploads bucket (versioning, SSE-S3, CORS enabled).
Create processed-images bucket (versioning, SSE-KMS).
Add folder structure: raw-uploads/ and processed/.

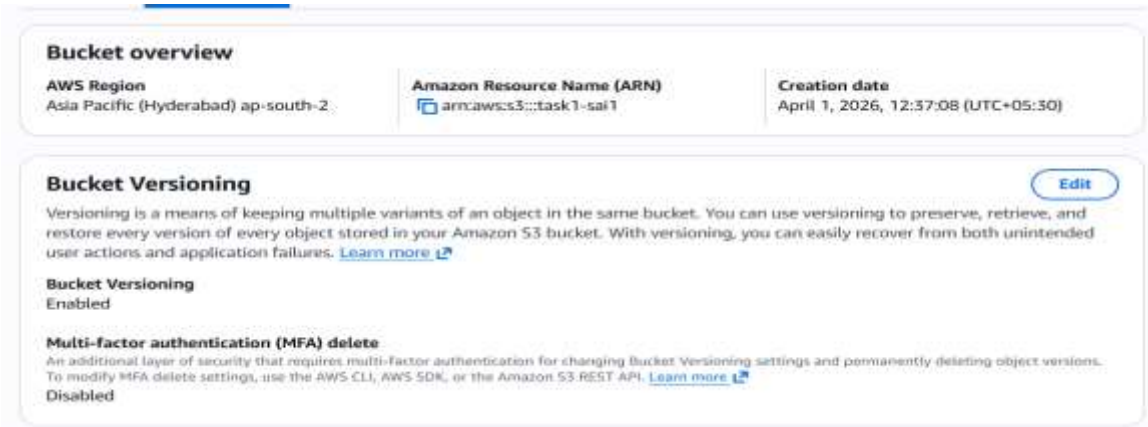


4. Cross-Region Replication

Enable replication from source bucket to eu-west-1.

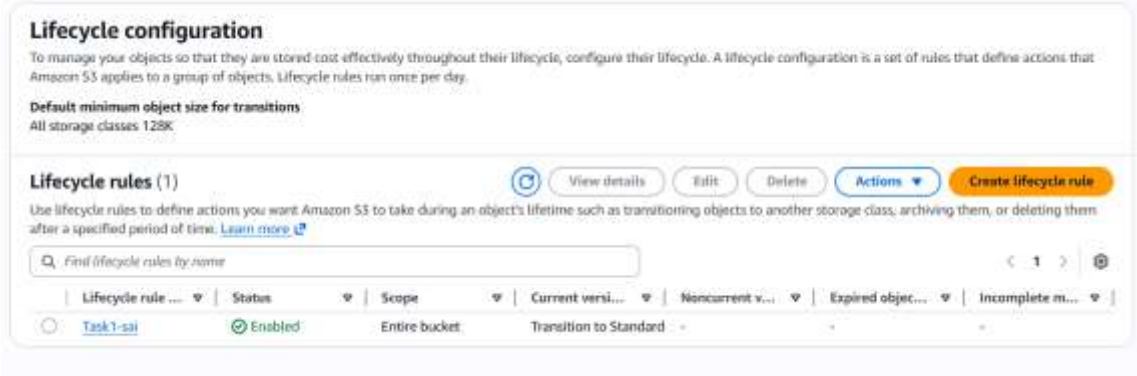


Ensure versioning is enabled on both buckets.



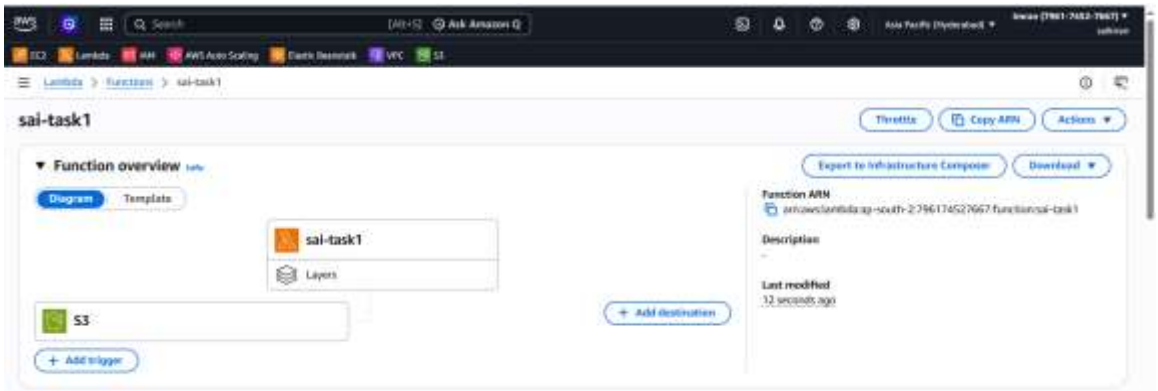
5. Lifecycle Rules

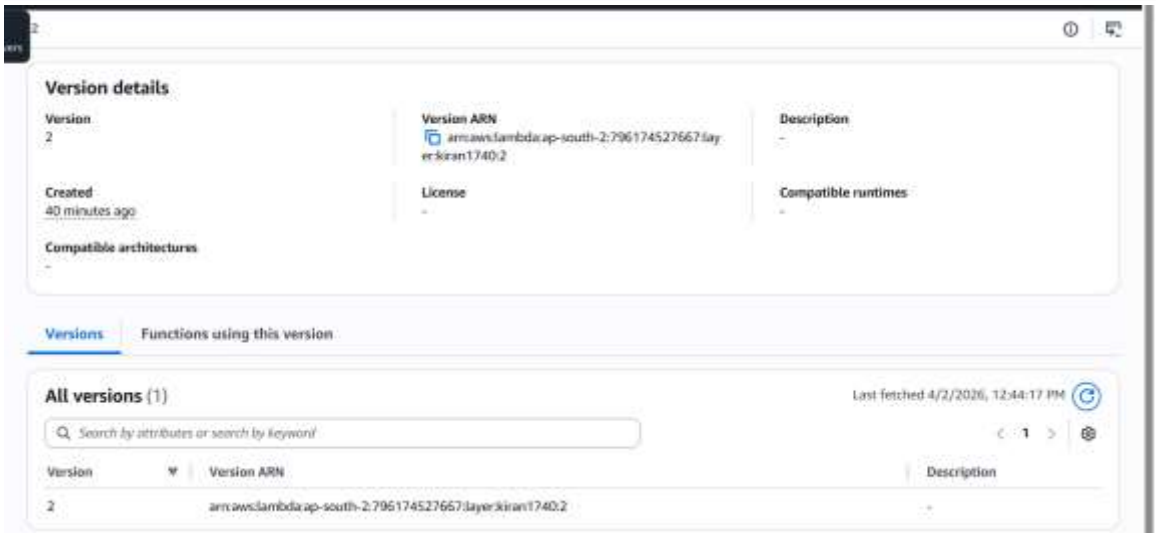
Configure lifecycle policy:
30 days → Standard-IA
365 days → Glacier



6. Lambda Image Processing

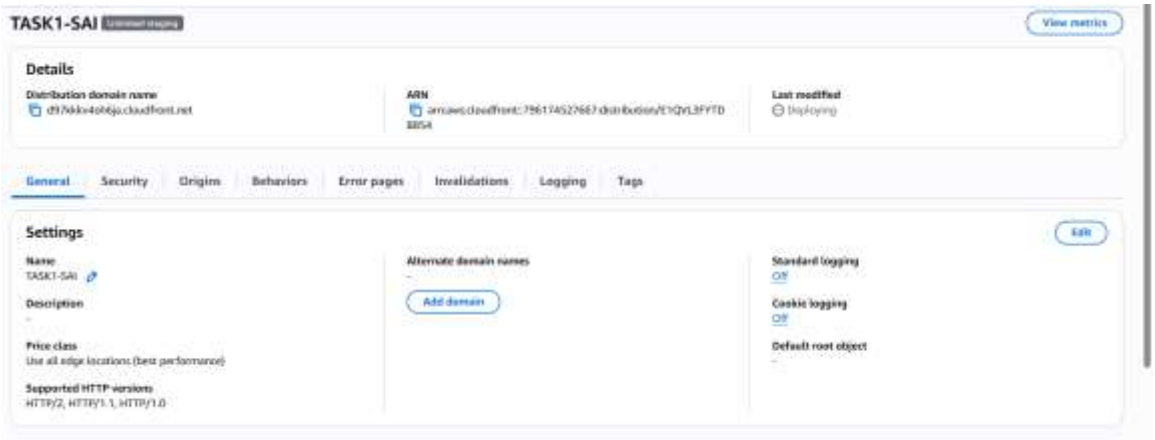
Create Lambda (Python 3.11, 512MB, 30 sec timeout).
Trigger: S3 ObjectCreated event.
Use Pillow layer for image processing.
Resize image to 800x600 and add watermark.





7. CloudFront Setup

Create distribution with processed bucket as origin.
Access images via CloudFront URL.



9. Conclusion

This project demonstrates a fully automated, scalable, and cost-efficient serverless pipeline on AWS.